

A1 Deep Q-Learning

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1 Task Description

The goal of the assignment was to train a Deep Q-Learning agent to navigate through the grid-based environment Minigrid-DoorKey6x6. The agent must pick up a key, unlock a door and reach the goal. The environment consists of 7 discrete actions and an symbolic observation of dimension (3, 7, 7).

2 Implementation

All models were trained locally on an NVIDIA GeForce RTX-4070 GPU.

The baseline was the exercise notebook provided. The first improvements were designated to reduce the loss and overfitting of the baseline model. Therefore, the learning rate was reduced to $4e-4$, MSE-Loss was replaced by SmoothL1Loss to lower the weight of the outliers and Gradient Clipping was introduced. Additionally, in order to have a meaningful training from the beginning, training starts only if there is a win in the Replaybuffer.

However, the greatest improvement was achieved by lowering the discount factor γ from 0.99 to 0.95 and increasing the number of training episodes to 2000. Locally, an average reward over 50 episodes of 0.97 was achieved. The reward-loss plot can be seen in 1. On the submission server a score of 0.95 was achieved.

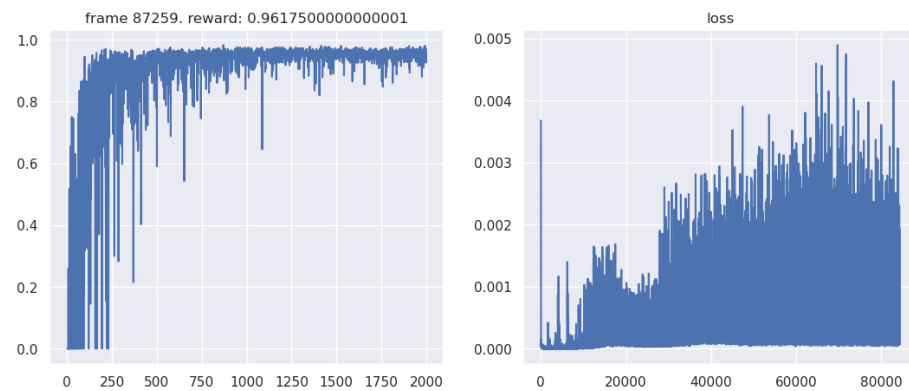


Figure 1: Reward-Loss plot of final model